BEHAVIORAL SYNTHETIC USERS: SIMULATING HOW REAL PEOPLE FAIL ON THE WEB

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Simulation Labs
BEHAVIORAL USER SIMULATION

Introduction

A funded category of "synthetic user research" interviews and surveys AI personas. Its documented flaw: the answers are **shallow and people-pleasing** — synthetic users claim they "finished all of it" where real users admit "three of seven."

The structural gap is simpler: **what people say and what people do are different things.** Attitudinal AI approximates the former; it cannot replicate the latter. Until computer-use models matured, simulating *doing* was impossible — now it isn't.

Aim

Simulate what users **do**, not what they say. Point a swarm of computer-use agents at any live website, give each a **mechanically imposed impairment**, and record exactly where — and why — each one abandons a real task (sign up, check out, cancel).

Methods

We exploit that H Company's **Holo** is a screenshot-in / action-out model: we degrade the pixels it perceives and the click it emits, *around* the model — mechanical fidelity, not roleplay.

Perturbation channels

- **Perception:** Gaussian blur (low vision), downscale (acuity), physiological CVD matrix (colour-blindness) — applied before inference, image dimensions preserved.
- **Actuation:** Gaussian coordinate noise (motor tremor) added *after* the model localizes.
- **Policy / viewport:** hard step + wall-clock budgets (impatience); a 390-px phone viewport.
- **Control:** an un-perturbed AI agent — is the site even **agent-ready**?

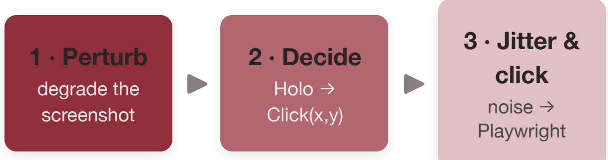

Figure 1. The per-step perceive–act loop, repeated until the task completes or the persona's budget runs out.

System

Five modules talk only through **frozen Pydantic contracts**, so the stack was built in parallel with zero merge conflicts. Async end-to-end; a swarm is **one Chromium launch, N isolated contexts**.

- **web → server:** a React live-grid and a FastAPI orchestrator — one async task per persona, streamed over WebSocket.
- **runner → engine:** Playwright drives the mouse by pixel; the engine perturbs the screenshot, calls Holo, and jitters the click.
- **report + voice:** survival curve & heatmap; Gradium cloned-voice exit-interviews.

Coordinates (verified live): Holo returns a 0–1000 grid; we denormalize to true pixels (x/1000·w) and execute verbatim — never rescaled.

Results

Representative run: `acme.com` sign-up, 8 personas, task = "create an account."

- **2 of 8 complete** — the control and the AI agent; every impaired persona abandons.
- **Differential failure:** personas die at distinct depths (Table 1), not at random.
- **Shared failure locus:** three personas abandon at the same pixel — the colour-only primary button.
- **Grounded, not guessed:** each abandonment carries a pixel, a video, and a voice.

Table 1. Progress reached & terminal outcome per persona.

#	Persona	Progress	Outcome
1	Alex (control)	100%	**success**
2	Agent (AI)	100%	**success**
3	Jordan (CVD)	74%	stuck
4	Luca (non-native)	61%	time budget
5	Sam (low vision)	47%	stuck
6	Dev (tremor)	38%	stuck
7	Margaret (72)	33%	step budget
8	Priya (impatient)	21%	time budget

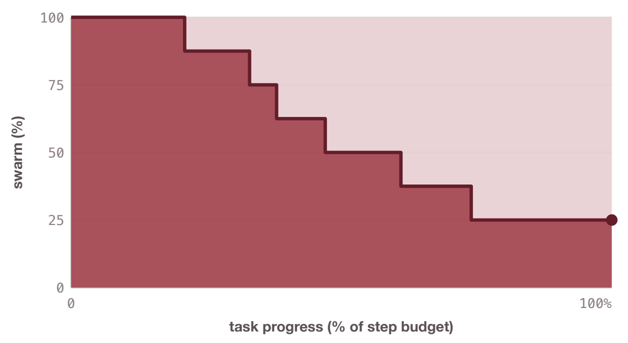

Figure 2. Swarm survival vs. task progress. Each downward step is one persona abandoning; the curve settles at the **25% completion plateau** (control + agent).

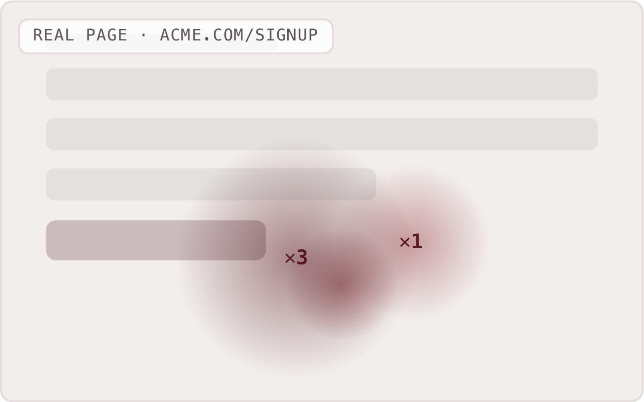

Figure 3. Abandonment heatmap over the real page. Deaths cluster on the colour-only password error and the ambiguous primary button — a concrete, fixable defect.

Benchmarks

Controlled evaluation — **is the impairment mechanical**, and does the pipeline localize correctly? Task = single-persona sign-up completion (n = 20 runs / condition).

Condition	Completion
Baseline — no perturbation	100%
Prompt only — "act low-vision"	90%
Mechanical — blur σ = 3.5	18%

Figure 5. Completion by condition. Prompting a persona to "act impaired" barely moves it; **degrading the input collapses it** — roleplay vs. simulation.

Click lands in target box — baseline	96%
…under tremor σ = 14 px	43%
Median localization error @ 1280×800	7 px
Seeded-defect recall (hostile form)	4 / 5

Discussion

The perturbation approach yields **mechanical fidelity**: we do not claim a persona *thinks* like a 72-year-old — only that the blur, coordinate noise and step budget are real perceptual and motor constraints on the model's own channels. Failures are therefore reproducible and tunable (e.g. tremor σ = 14 px).

Because the swarm must *complete* the task, the people-pleasing failure mode cannot occur — an agent either got through the form or it did not. **Positioning:** Tester H checks whether software works *as specified*; Simulation Labs checks whether **humans survive it** — upstream and complementary.

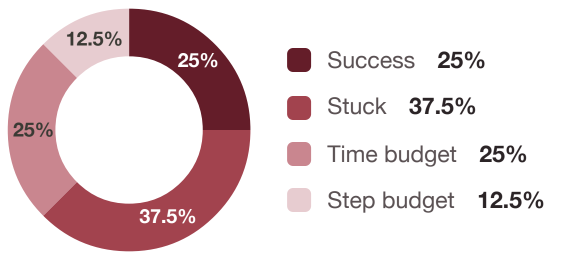

Figure 4. Terminal-outcome distribution across the swarm. Only *success* counts as completion; the rest abandon by a distinct mechanism.

Conclusion

Simulation Labs reframes synthetic user research from **attitudinal to behavioral**. Applications: accessibility-compliance evidence (ADA; EU Accessibility Act, 2025), pre-launch funnel forensics, and **agent-readiness certification**. Voice exit-interviews (Gradium) turn each abandonment into empathy; a NemoClaw policy sandbox lets the swarm browse but never submit or pay.

Point it at your product before launch, and hear your users fail — before they're real.

References

1. Nielsen Norman Group. *Synthetic Users: When & How to Use AI-Generated Research.* 2024.
2. Machado, Oliveira & Fernandes. *A Physiologically-based Model for Simulation of Colour Vision Deficiency.* IEEE TVCG, 2009.
3. H Company. *Holo 3.1* computer-use model (`holo3-1-35b-a3b`).
4. Gradium — real-time TTS/STT & voice cloning. NVIDIA *NemoClaw / OpenShell* policy sandbox.

Scan for code, demo & receipts
Full source, the 90-second demo playbook, and sample video/voice artifacts. `github.com/SatyamDave/simulation-labs`